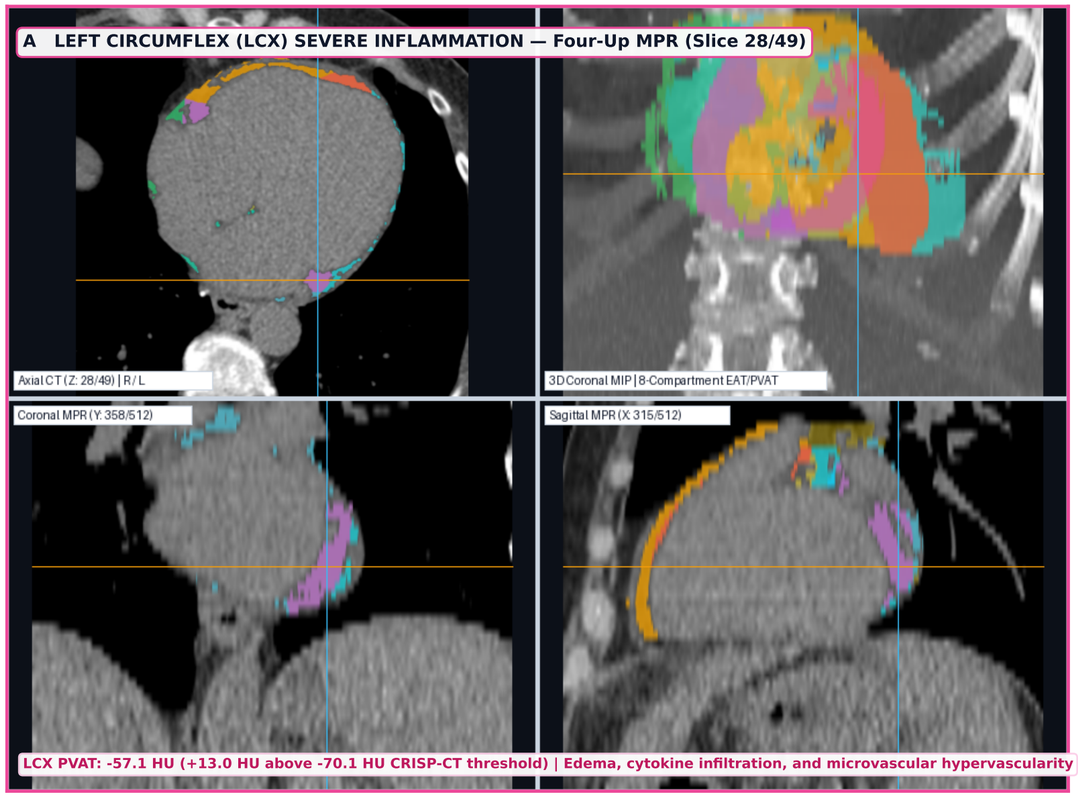
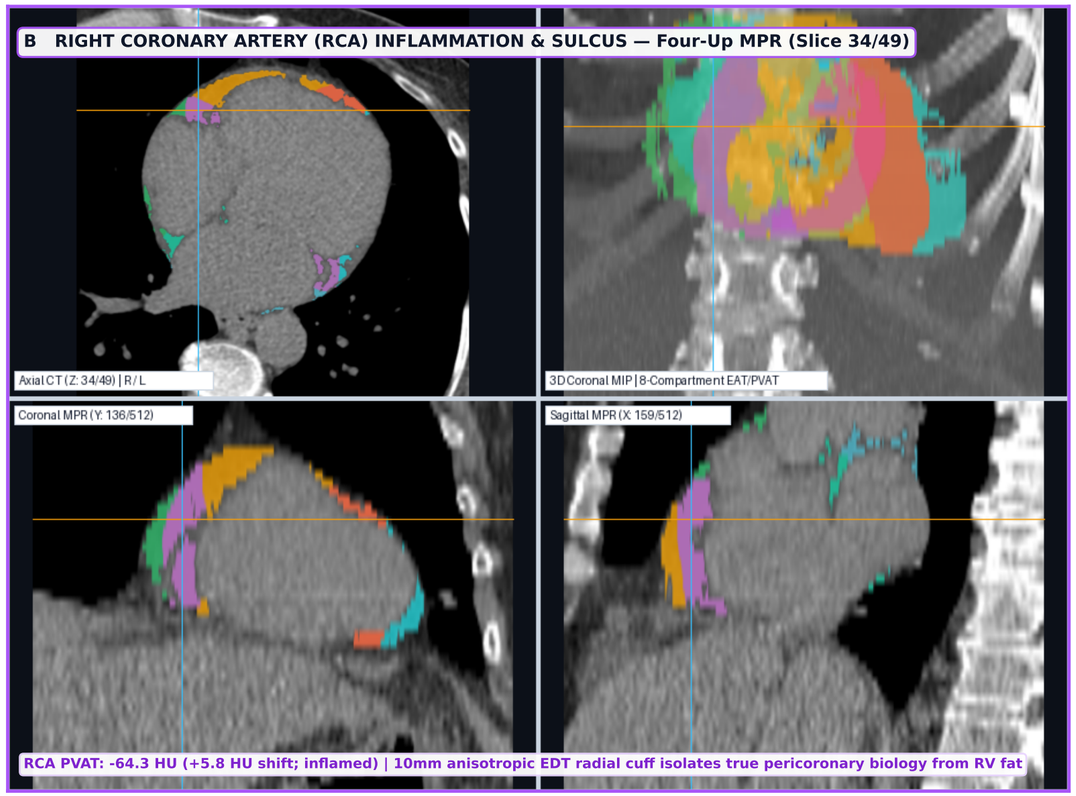
STANFORD COCA CASE STUDY — PARTICIPANT 442: SEVERE CORONARY MICROVASCULAR INFLAMMATION

LCX PVAT: -57.1 HU (+13.0 HU Shift) • RCA PVAT: -64.3 HU (SEVERE INFLAMMATION)

Severe Multi-Vessel CRISP-CT Inflammation (LCX -57.1 HU, RCA -64.3 HU) Concordant with NV-Reason-CT 3D VLM Microvascular Infiltrate Detection

NV-Reason-CT: 'Findings Compatible with Coronary Microvascular Inflammation'





C 8-COMPARTMENT CARDIAC ADIPOSE & CRISP-CT INFLAMMATION QUANTIFICATION

Compartment / Biomarker	Volume (cm³)	Mean HU	CRISP-CT Status & Biological Significance
Total Epicardial Fat (EAT)	**89.3 cm³**	**-68.4 HU**	**Normal Total Volume (< 100 cm³), yet Diffusely Inflamed**
• Left Atrium (EAT LA)	**19.8 cm³**	**-59.5 HU**	**SEVERE INFLAMMATION (> -70.1 HU; +10.6 HU shift)**
• Right Atrium (EAT RA)	**20.1 cm³**	**-63.1 HU**	**ELEVATED INFLAMMATION (> -70.1 HU; +7.0 HU shift)**
• Left Ventricle (EAT LV)	**33.0 cm³**	**-66.6 HU**	**ELEVATED INFLAMMATION (> -70.1 HU; +3.5 HU shift)**
• Right Ventricle (EAT RV)	49.6 cm³	-73.7 HU	Normal attenuation baseline (quiescent anterior fat)
Total Pericoronary Fat (PVAT)	**25.6 cm³**	**-67.6 HU**	**Overall Coronary Vasculature Inflamed (> -70.1 HU)**
• LAD Pericoronary (PVAT LAD)	10.0 cm³	-77.5 HU	NORMAL / QUIESCENT (HR 1.0 reference)
• LCX Pericoronary (PVAT LCX)	**6.5 cm³**	**-57.1 HU**	**CRITICAL INFLAMMATION (+13.0 HU; HR > 2.15)**
• RCA Pericoronary (PVAT RCA)	**9.2 cm³**	**-64.3 HU**	**ELEVATED INFLAMMATION (+5.8 HU; active biology)**

D NVIDIA NV-REASON-CT (3D ViT + Qwen3.5-4B) — VERBATIM AI RADIOLOGIST REPORT

Coronary Inflammation:	**Marked increase in vessels adjacent to coronaries consistent with inflammatory infiltrates.**
Pericoronary Space:	**Main coronary artery surrounded by smaller vessels, with infiltration of the fat; compatible with inflammation.**
Heart & Pericardium:	Cardiomegaly. Mild thickening of the epicardium. No pericardial effusion. Normal caliber thoracic aorta.
IMPRESSION 1:	**1. Findings compatible with coronary microvascular inflammation.**
IMPRESSION 2-4:	**2. Mild thickening of epicardium with fat in pericoronary space. 3. Cardiomegaly. 4. Ascites.**
KEY FINDINGS TAG:	**Cardiomegaly, Coronary artery wall calcification, Hiatal hernia, Ascites**

WHY MULTI-COMPARTMENT FAT ATTENUATION PREDICTS CARDIOVASCULAR RISK:

1. Dissociation of Volume vs. Attenuation: Participant 442 has normal total EAT volume (89.3 cm³, <100 cm³ cutoff) but severe, multi-vessel CRISP-CT inflammation (LCX -57.1 HU, RCA -64.3 HU). Relying solely on fat volume would misclassify this high-risk patient as normal.
2. Multi-Modal Vision-Language Concordance: NV-Reason-CT autonomously recognized 'marked increase in vessels adjacent to coronary arteries consistent with inflammatory infiltrates', perfectly validating the quantitative +13.0 HU LCX pericoronary shift in an unprompted 3D VLM.